# Questions from Tutorial 3

6. Gegee die volgende frekwensieverdeling vir $n = 100$ : /
*Given the following frequency distribution for* $n = 100$

| Klasgrense<br>*Class limits* | Frekwensie<br>*Frequency* |
|---|---|
| $(0;10]$ | 40 |
| $(10;20]$ | 25 |
| $(20;30]$ | 15 |
| $(30;40]$ | 15 |
| $(40;50]$ | 5 |
| Totaal / *Total* | 100 |

   a. Benader die gemiddelde. / *Approximate the mean.*
   b. Benader die standaardafwyking. / *Approximate the standard deviation.*

18. Is die punte wat 'n student vir Statistiek 188 behaal verwant aan die punt wat hy/sy vir Wiskunde in matriek verwerf het? Om hierdie moontlikheid te ondersoek, neem 'n navorser 'n ewekansige steekproef van tien studente wat ingeskryf was vir Statistiek 188. /
*Are the marks a student obtains for Statistics 188 related to the marks they obtained for Mathematics in matric? To analyze this possibility a researcher takes a random sample of ten students who had enrolled for Statistics 188.*

Die studente se finale punte vir Wiskunde en Statistiek 188 is as volg: /
*The students' final marks for Mathematics and Statistics 188 are as follows:*

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Wiskundepunt /** ***Mathematics Mark*** | 89 | 73 | 57 | 53 | 51 | 49 | 47 | 44 | 42 | 38 |
| **Statistiekpunt /** ***Statistics Mark*** | 51 | 53 | 49 | 50 | 48 | 21 | 46 | 19 | 43 | 43 |

   a. Bereken en interpreteer die kovariansie. /
   *Calculate and interpret the covariance.*
   b. Bereken en interpreteer die korrelasiekoëffisiënt. /
   *Calculate and interpret the coefficient of correlation.*

# Questions from Tutorial 4

38. Beskou twee gebeurtenisse, $A$ en $B$ in 'n steekproefruimte $S$. Die volgende waarskynlikhede word gegee: / *Consider two events, A and B in a sample space S. The following probabilities are given:*

$$P(A) = 0.3;\ P(A' \text{ en/}and\ B) = 0.22;\ P(B \mid A) = 0.6$$

Bereken: / *Calculate:*
(a) $P(A$ en/*and* $B)$
(b) $P(B)$
(c) $P(A \mid B)$
(d) $P$(minstens een van $A,B$) / *P(at least one of A,B)*
(e) $P$(presies een van $A,B$) / *P(exactly one of A,B)*

# Questions from Tutorial 5

17. In 'n besending van 15 hardeskywe is vyf defektief. Indien vier van die skywe inspekteer word, wat is die waarskynlikheid dat: / *In a shipment of* 15 *hard disks, five are defective. If four of the disks are inspected, what is the probability that*:

(a) Presies een defektief is? / *Exactly one is defective*?
(b) Ten minste een defektief is? / *At least one is defective*?
(c) Nie meer as twee defektief is nie? / *No more than two are defective*?
(d) Wat is die gemiddelde aantal defektiewe hardeskywe wat u sou verwag om te vind in 'n steekproef van vier hardeskywe? / *What is the mean number of defective hard disks that you would expect to find in the sample of four hard disks*?

# Questions for normal distribution

| | |
|---|---|
| 7. Consider a normal distribution $X$ with a mean of 100 and a standard deviation of 15. Answer the questions that follow. | 7. Beskou 'n normaalverdeling $X$ met 'n gemiddeld van 100 en 'n standaardafwyking van 15. Beantwoord die vrae wat volg. |
| (a) Calculate $P(X > 88.45)$. | (a) Bereken $P(X > 88.45)$. |
| (b) Find the value $x$ such that $P(X > x) = 0.1736$. | (b) Bereken die waarde $x$ sodat $P(X > x) = 0.1736$. |
| (c) Find the two values $a$ and $b$, symmetric around the mean of $X$, such that $P(a < X < b) = 0.95$. | (c) Vind die twee waardes $a$ en $b$, simmetries rondom die gemiddelde van $X$, sodat $P(a < X < b) = 0.95$. |
| (d) Calculate $P(X < 115 \mid X > 108.1)$. | (d) Bereken $P(X < 115 \mid X > 108.1)$. |